

Task 3 - Regulatory Mapping Table

Jurisdiction-Aware Credit Decision Governance Monitor | US and EU rule-to-tool mapping

This mapping table is the legal-policy backbone of the prototype. It shows that the US and EU modules do not merely use different labels. They have different data permissions, computation modules, outputs and human owners.

Jurisdiction	Regulatory concern	Legal / policy basis	Tool response	Data access rule	Output generated	Human owner	Limitation
US	Specific adverse action reasons for declined or unfavourably treated applicants	ECOA / Regulation B 12 CFR 1002.9; CFPB Circular 2022-03	Use SHAP-style attribution only as an internal aid, then map to approved reason codes and flag low-confidence cases.	No protected attributes in US live explanation path.	Draft adverse action evidence package and sampled review queue.	US Compliance Lead	Tool does not issue the final legal notice.
US	Proxy discrimination and digital redlining risk	ECOA fair lending risk management; CFPB public guidance on AI and digital redlining	Monitor postcode, thin-file and income-instability concentration as proxy-risk signals.	Live path uses non-protected variables; demographic analysis is separated into privileged review if approved.	Fair lending internal review trigger.	Fair Lending Review Counsel	Without direct demographic labels, proxy testing can under-detect harm.
US	Sensitive-data contamination in live credit tooling	Internal policy boundary aligned with data minimisation and fair lending controls	Physically hard-drop audit-only fields before US processed data enters the tool.	US processed file excludes age, gender_audit, age_band, protected_group_flag and audit_group.	Data contract pass/fail control.	Data Governance Lead	Does not remove the need for separately approved fair lending analysis.
EU	Creditworthiness AI as high-risk AI	EU AI Act Annex III for creditworthiness / credit score high-risk classification.	Default EU credit decision use case into high-risk AI governance workflow.	Governance checklist does not require live sensitive-data use.	Pre-deployment evidence completeness gate.	EU AI Governance Lead	Tool does not provide conformity certification.
EU	Bias testing and lifecycle risk management	EU AI Act high-risk AI obligations, including data governance, risk management and human oversight. Sensitive or special-category audit fields require a separate lawful basis, access control, logging, minimisation and retention controls.	Run group performance and approval disparity tests in segregated audit environment.	Sensitive or special-category fields only in audit environment with lawful basis, access control, logs, minimisation and retention controls.	Bias testing evidence package.	EU Model Risk Lead	Synthetic audit labels demonstrate method, not real protected-class prevalence.
EU	Post-deployment monitoring and material change	EU AI Act high-risk AI lifecycle governance and post-deployment / post-market monitoring expectations.	Monitor PSI drift, reason-code instability and performance shift by audit group.	Production monitoring data plus audit-only comparison where approved.	Drift ticket and model governance review trigger.	Model Monitoring Lead	Drift detection does not identify every causal source of model degradation.